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#include <Wire.h>
#include <Adafruit_GFX.h>
#include <Adafruit_SSD1306.h>
#include <DHTesp.h>
#define SCREEN_WIDTH 128
#define SCREEN_HEIGHT 64
#define PIR_PIN 13
#define GREEN_LED 2
#define RED_LED 4
#define BLUE_LED 5
#define DHT_PIN 15
Adafruit_SSD1306 display(SCREEN_WIDTH, SCREEN_HEIGHT, &Wire, -1);
DHTesp dhtSensor;
void setup() {
  Serial.begin(115200);
  pinMode(PIR_PIN, INPUT);
  pinMode(GREEN_LED, OUTPUT);
  pinMode(RED_LED, OUTPUT);
  pinMode(BLUE_LED, OUTPUT);
  dhtSensor.setup(DHT_PIN, DHTesp::DHT22);
  if(!display.begin(SSD1306_SWITCHCAPVCC, 0x3C)) {
    Serial.println("OLED Failed");
    while(true);
  }
  display.clearDisplay();
  display.setTextSize(1);
  display.setTextColor(SSD1306_WHITE);
  display.setCursor(15,20);
  display.println("SMART HOSTEL");
  display.display();
  delay(2000);
}
void loop() {
  int motion = digitalRead(PIR_PIN);
  TempAndHumidity data = dhtSensor.getTempAndHumidity();
  float temp = data.temperature;
  float hum = data.humidity;
  String roomStatus;
  String fanStatus;
  if(motion == HIGH) {
    roomStatus = "OCCUPIED";
    digitalWrite(GREEN_LED, HIGH);
    digitalWrite(RED_LED, LOW);
    if(temp > 30) {
      digitalWrite(BLUE_LED, HIGH);
      fanStatus = "ON";
    }
  }
  else {
    digitalWrite(BLUE_LED, LOW);
    fanStatus = "OFF";
  }
}

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}
else {
    roomStatus = "EMPTY";
    fanStatus = "OFF";
    digitalWrite(RED_LED, HIGH);
    digitalWrite(GREEN_LED, LOW);
    digitalWrite(BLUE_LED, LOW);
}
// OLED Display
display.clearDisplay();
display.setCursor(0,0);
display.println("SMART HOSTEL");
display.setCursor(0,15);
display.print("Room: ");
display.println(roomStatus);
display.setCursor(0,30);
display.print("Temp: ");
display.print(temp);
display.println(" C");
display.setCursor(0,45);
display.print("Fan: ");
display.println(fanStatus);
display.display();
// Serial Log
Serial.println("=====");
Serial.print("Room Status : ");
Serial.println(roomStatus);
Serial.print("Temperature : ");
Serial.print(temp);
Serial.println(" C");
Serial.print("Humidity : ");
Serial.print(hum);
Serial.println(" %");
Serial.print("Fan Status : ");
Serial.println(fanStatus);
Serial.println("=====");
delay(2000);
}

```